

Final Project — Notes on Readings

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1 Literature Group A: Algorithm Structure & Theory

1.1 Johnson, Logistic Matrix Factorization for Implicit Feedback Data (2014)

1.1.1 Motivation

- Recommender systems come in two categories: content based and collaborative filtering.
- **Content based:** User age, sex, and other demographics used to associate products
- Collaborative filtering with **implicit feedback data:** recommender system techniques that analyze relationships between users and items using implicit signals (e.g., CTR, play counts) to provide personalized recommendations
- Collaborative filtering with **explicit feedback data:** models these relationships using explicit signals (e.g., ratings)
- **User-oriented neighborhood methods** search the domain of U to find users that have rated or interacted with similar items and then recommend items that similar users prefer. **Item-oriented neighborhood methods** instead search the domain of I to find items similar to the items that a user has either rated highly or interacted with. **Matrix Factorization models** (part of latent factor model family) seek to find a low rank approximation to a sparse ratings matrix by minimizing a loss function on the known ratings
- **Logistic Matrix Factorization (Logistic MF):** a probabilistic model for matrix factorization with implicit feedback. Can model probability of a user preferring a specific item, and outperforms Implicit Matrix Factorization

1.1.2 Setup

- Implicit feedback can come in many forms (e.g., clicks, views, or streams), but in all cases, assume we have a set of non-negative feedback values associated with each pair of users and items in domain:
 - $U = (u_1, \dots, u_n)$: group of n users
 - $I = (i_1, \dots, i_m)$: group of m items
 - $R = (r_{ui})_{n \times m}$: user-item observation matrix where $r_{ui} \in \mathbb{R}_{\geq 0}$ represents number of times user u interacted with item i .
- r_{ui} values not integers so they can be textually or temporally weighted (e.g., stream with click & stream without click). 0 does not mean non-preferred—could mean unknown to user

- Goal is to find top recommended items for each user for each item they have not yet interacted with (i.e., $r_{ui} = 0$)

1.1.3 Logistic MF

- Factorizes an extremely sparse user-item observation matrix into two lower-dimensional matrices representing latent factors for users and items.
- The probability that user u prefers item i is modeled using a logistic function.
- The equation integrates user and item biases to account for behavioral variations, such as highly active users or globally popular items.
- **Formal Equation:** The probability of preference is defined as:

$$p(l_{ui}|x_u, y_i, \beta_u, \beta_i) = \frac{\exp(x_u y_i^T + \beta_u + \beta_i)}{1 + \exp(x_u y_i^T + \beta_u + \beta_i)}$$

- Non-zero observations are treated as positive feedback and assigned a confidence weight proportional to the number of interactions, while missing data (zeros) are treated as single negative observations.
- Zero-mean spherical Gaussian priors are placed on the latent vectors, effectively functioning as L_2 regularization to prevent overfitting.

1.1.4 Training and Scalability

- **Optimization:** The model maximizes the log posterior using an alternating gradient ascent procedure.
- **Adaptive Learning:** Utilizes AdaGrad to adaptively adjust gradient step sizes during training, reducing the iterations required for convergence.
- **Parallelization (MapReduce):** The observation matrix and latent vectors are sharded into smaller blocks so they can fit into memory. Computations are mapped to specific users and items, then efficiently aggregated during the reduce phase.

1.1.5 Experimental Results

- Evaluated against a popularity baseline and the Implicit Matrix Factorization (IMF) model using a Spotify dataset consisting of 50,000 users and 10,000 artists over two years.
- **Evaluation Metric:** Performance was measured using Mean Percentage Ranking (MPR), a recall-based metric where lower scores indicate that the user listened to artists higher in their predicted lists.
- **Accuracy:** Logistic MF achieved an MPR of 5.99% at 100 latent factors, successfully outperforming both the IMF model (6.27%) and the global popularity baseline (14.9%).
- **Efficiency with Fewer Factors:** A major advantage was Logistic MF's ability to achieve high accuracy using far fewer latent dimensions. At 10 latent factors, Logistic MF achieved an MPR of 8.065%, whereas IMF scored 16.9%.

- **Production Impact:** Requiring fewer latent factors is a significant operational advantage, as it drastically reduces the RAM required to maintain the complex data structures (like k-d trees or LSH-based trees) used for real-time approximate nearest neighbor (ANN) searches.

1.2 Peng et al., Reconciling the Accuracy-Diversity Trade-off in Recommendations (2024)

1.2.1 Motivation

- **The Apparent Trade-off:** Traditional recommender systems face a tension between *accuracy* (showing items a user is most likely to want) and *diversity* (showing a range of categories). Injecting diversity heuristically usually artificially lowers measured accuracy.
- **Core Argument:** The paper argues that accuracy misrepresents true **user utility** because it ignores **consumption constraints**. At any given time, users typically only consume one or a few items (e.g., watching a single movie or eating at one restaurant).
- When optimizing for a precise measure of utility (the probability that a user likes *at least one* recommended item), recommendations naturally become diverse due to the diminishing returns of recommending similar items.

1.2.2 Setup and Objectives

- Assume m discrete types of items (e.g., movie genres). A user is in the mood for type t with probability p_t .
- Conditional on preferring type t , the user likes the i -th item of type t with probability $q_{t,i}$.
- In the general model, item quality decays as more items of a type are recommended. This is parameterized as $q_{t,i} = q_t(i + \beta)^{-\alpha}$, where q_t is the relative quality of the type as a whole, and α is the rate of decay.
- **Accuracy Objective (acc):** The expected proportion of items in the recommendation set S that the user likes.
- **Utility Objective ($util_1$):** The probability that the user likes *at least one* item in S (unit consumption constraint).

1.2.3 Main Results and Theorems

- **Theorem 1 (Basic Setting):** When item quality is constant across all types ($q_{t,i} = q$):
 - Accuracy-maximizing recommendations are 100% homogeneous, representing only the single type the user prefers most frequently ($\arg \max p_t$).
 - Utility-maximizing recommendations become completely diverse asymptotically, representing an equal proportion ($1/m$) of items from each type, completely ignoring the base probabilities (p_t).
- **Theorem 2A & 2B (General Setting with Decay α):**
 - **Low Decay ($\alpha < 1$):** The system has a deep catalog of high-quality items per category. Accuracy remains highly homogeneous, while utility diversifies heavily to cover all possible user moods. The “accuracy-diversity trade-off” is most severe here.

- **High Decay ($\alpha > 1$):** The system quickly runs out of good items in a given category. Both accuracy and utility naturally lead to highly diverse recommendations, dissolving the trade-off.
- **Corollary 3 (The “Milk and Ice Cream” Theorem):** Shows that an item’s representation in a recommendation set should vary inversely with its conditional quality q_i . A platform should recommend *more* items from categories where the individual items have a *lower* likelihood of satisfying the user.
 - *Example:* A customer is more likely to buy milk than ice cream ($p_{milk} > p_{ice_cream}$), but a specific ice cream flavor has a lower conditional satisfaction rate than generic milk ($q_{ice_cream} < q_{milk}$). To maximize the chance the user finds something they want, the system must dedicate significantly more “shelf space” to ice cream.

1.3 Wang et al., Causal Inference for Recommender Systems (2020)

1.3.1 Motivation

- The task of recommender systems is classically framed as predicting users’ preferences and ratings.
- However, recommendation is fundamentally a causal inference question: “What would the rating be if we ‘forced’ the user to watch the movie?”
- The key challenge is unobserved confounders—variables that affect both the treatment assignment (which items users decide to interact with) and the outcome (how they rate them).
- Traditional matrix factorization only provides valid causal inferences if users interacted with items purely at random.

1.3.2 Setup and Causal Framework

- The paper recasts recommendation using potential outcomes notation from the Rubin causal model.
- Let a_{ui} be the binary indicator of whether user u rated item i (the exposure or treatment).
- Let $y_{ui}(1)$ be the potential rating user u would give if exposed to item i . Let $y_{ui}(0)$ be the potential rating if they do not see it.
- Classical causal inference requires *strong ignorability* (the assumption that all confounders are measured and controlled for), which is unfortunately untestable.

1.3.3 The Deconfounded Recommender

- The proposed algorithm leverages the multiplicity of treatments (each user is exposed to many movies) to adjust for unobserved confounders. It involves three main steps:
- **1. Model the Exposure:** Fit a model to the exposure data (who watched what) to understand discovery patterns. The authors use Poisson Factorization (PF) for this step.
- **2. Compute Substitute Confounders:** Using the fitted PF model, compute the posterior predictive mean to reconstruct the exposure matrix: $\hat{a}_{ui} = \mathbb{E}_{PF}[\pi_u^\top \lambda_i | a]$. This $\hat{a}_{ui}$ serves as a substitute confounder.

- **3. Fit the Outcome Model:** Fit a classical ratings model that incorporates the substitute confounder:

$$y_{ui}(a) = \theta_u^\top \beta_i \cdot a + \gamma_u \cdot \hat{a}_{ui} + \epsilon_{ui}$$

- γ_u is a user-specific coefficient describing how much the substitute confounder contributes to the rating, allowing the model to estimate unbiased preferences (θ_u, β_i) .

1.3.4 Experimental Results

- Evaluating causal models requires a random test set (where users rate randomly assigned items) to act as an unbiased estimate of performance across all potential outcomes.
- Tested on simulated data and real-world datasets with random test sets (Yahoo! R3 and Coat shopping).
- The deconfounded recommender outperformed classical counterparts (Probabilistic MF, Poisson MF, and Weighted MF).
- It also outperformed the existing causal baseline, Inverse Propensity Weighting (IPW) Matrix Factorization.
- The improvements were consistent for both weak generalization (predicting unseen items for existing users) and strong generalization (predicting preferences for entirely new users).

2 Literature Group B: User Behavior & Serendipity

2.1 Anderson et al., Algorithmic Effects on the Diversity of Consumption on Spotify

2.1.1 Core Question

Do recommendation algorithms make users' listening narrower or broader over time, and does it matter?

2.1.2 Method

Analyzed 100M+ Spotify premium users over July 2019. Used a "generalist-specialist score" (GS-score), a song embedding where similar songs cluster together, to measure how musically diverse each user's listening is. Distinguished between organic (user-driven: search, own playlists) and programmed (algorithm-driven: Discover Weekly, radio) listening.

2.1.3 Key Findings

- Algorithmic/programmed listening is consistently less diverse than organic listening
 - Recommendations push users toward similar content
- Musical diversity is surprisingly stable over time (correlation of 0.56 over one year), suggesting it reflects a fundamental behavioral trait
- Users who become more diverse do so by shifting toward organic listening and away from algorithmic content
- Higher diversity strongly predicts better long-term outcomes
 - Generalists are up to 25pp less likely to churn and up to 35pp more likely to convert to paid subscribers
 - **NOTE:** pp means percentage points
- In an A/B experiment on "All Out" decade playlists (540K users), relevance-based ranking worked much better for specialists than generalists
 - Specialists got a 25.7% stream boost vs. 10% for generalists, suggesting current recommendation approaches are poorly suited to diverse listeners

2.1.4 NOTE

Central Tension this Paper Identifies: Algorithms are effective at short-term engagement but may suppress the diversity that actually drives long-term retention. Optimizing for relevance and optimizing for user health may be in conflict.

2.2 Mok et al., The Dynamics of Exploration on Spotify

2.2.1 Core Question

How do users explore new music over time, and how does this change with age? Is exploration the same thing as diversity?

2.2.2 Method

Longitudinal study of 100K US Spotify premium users over 4 years (2016–2019). Defined exploration as discovering new tracks (first-time listens) and turnover (how much a user's weekly listening set changes). Compared these to the GS-score diversity metric from Anderson et al.

2.2.3 Key Findings

- Older users explore more, younger users exploit more
 - Older listeners consistently discover more new tracks and turn over more of their weekly listening
 - Younger listeners re-listen to the same songs more heavily
- Younger users do convert their discoveries into favorites faster (they lock onto new songs more quickly once found)
- Exploration happens in bursts/phases, not at a steady pace
 - If you're in an exploratory week, the next week will likely also be exploratory (chaining effect)
- There are strong seasonal patterns: exploration spikes around Christmas, especially for older users (holiday music), then drops sharply
- **Exploration and diversity are largely independent** ($r \approx 0.15 - 0.19$)
 - Younger users are more diverse despite exploring less
 - Older users explore more but are less diverse
 - The paper coins useful archetypes:
 - * "Exploiter-generalists" (young, diverse but not exploratory) vs.
 - * "Explorer-specialists" (old, explorative but narrow)

2.2.4 NOTE

Relevance for Literature Review: It complicates Anderson et al. Diversity and exploration capture different dimensions of heterogeneous consumption. A recommender system that increases one won't necessarily increase the other.

2.3 Kotkov et al., A Survey of Serendipity in Recommender Systems

2.3.1 Core Question

What is serendipity in recommender systems, how is it measured, and how can algorithms produce it?

2.3.2 Conceptual Framework

Serendipity = **relevant** + **novel** + **unexpected**

- *Relevant*: the user actually likes it

- *Novel*: the user has never encountered it
- *Unexpected*: it differs significantly from the user's existing profile (not just unfamiliar, genuinely surprising)
- Serendipity is not the same as diversity and not the same as novelty
 - Diversity compares recommended items to each other
 - Serendipity compares them to the user's profile
 - Novelty doesn't require unexpectedness or relevance

2.3.3 Why Serendipity is Hard

- Users get bored with overly safe suggestions (the "overspecialization problem") but serendipitous items are risky
 - Most items that are unexpected are also irrelevant
- The concept has an emotional/subjective dimension that's hard to formalize
- No consensus definition existed at time of writing

2.3.4 Algorithmic Approaches Surveyed

- *Reranking*: run a normal accuracy-based algorithm, then rerank results to penalize obvious suggestions (e.g., Full Auralist, Utility Model)
- *Modifications*: tweak standard algorithms (like k-nearest neighbor) to use "furthest neighbors" or unexpected similarities instead of closest ones
- *Novel algorithms*: approaches like TANGENT, which recommends items liked by users from different taste groups than the target user

2.3.5 Evaluation Challenges

- Offline metrics (using historical data) are poor proxies for serendipity because datasets don't record what users would have been surprised by
- Online user studies work better but are expensive
 - Even then, users often don't know what "serendipitous" means when asked directly
- Most metrics use a "primitive recommender" as a baseline and measure how much the serendipitous algorithm deviates from it (
 - Sensitive to what baseline you choose

2.3.6 NOTE

Future Directions this Paper Identifies: Context-aware serendipity (mood, time of day), cross-domain recommendations, group recommendations.

2.4 Cross-Paper Themes

1. The overspecialization problem runs through all three papers.
 - (a) Anderson et al. documents it empirically on Spotify.
 - (b) Kotkov et al. frames it as the central problem serendipity research is trying to solve.
 - (c) Mok et al. shows it's even more nuanced once you separate exploration from diversity.
2. Diversity, exploration, and serendipity are related but distinct constructs.
 - (a) Mok et al. makes this most explicit, but it's implicit throughout.
 - (b) The Group B papers are essentially mapping out the conceptual space of "heterogeneous consumption."
3. Short-term vs. long-term tension.
 - (a) Anderson et al. quantifies it most directly: relevance-optimized algorithms win on immediate engagement but correlate with worse long-term retention.
 - (b) Kotkov et al. addresses it through the accuracy-serendipity tradeoff.
 - (c) This tension likely connects directly to Group A — particularly Wang (causal inference) and Peng (accuracy-diversity tradeoff), which probably tackle it from the algorithm design side.
4. Measurement is hard and consequential (each paper wrestles with how to measure its central concept).
 - (a) GS-score (Anderson, Mok) is content-aware but coarse.
 - (b) Exploration metrics (Mok) are granular but content-agnostic.
 - (c) Serendipity metrics (Kotkov) are sensitive to baseline choice.